

## Algebra First-Year Exam, August 2022, Part 1

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Answer 4 questions. Write your solutions clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Prove that the alternating group  $A_n$ ,  $n \geq 3$ , is generated by 3-cycles. (You may assume that  $S_n$  is generated by transpositions.)
2. This problem has two parts.
  - (a) Prove that a group  $G$  of order 75 has a unique subgroup  $H$  of order 25.
  - (b) Show that if the above subgroup  $H$  is cyclic then  $G$  is abelian.
3. Let  $G$  be symmetric group  $S_n$  acting on  $X = \{1, 2, \dots, n\}$ ,  $n \geq 2$ .
  - (a) Explain how we have an induced action on the set  $X \times X$  of ordered pairs  $(a, b)$ , with  $a, b \in \{1, 2, \dots, n\}$ .
  - (b) Show that there are two orbits of the action on  $X \times X$  and describe them.
  - (c) Describe the stabilizer in  $G$  of an element in each of the two orbits.
4. This problem is about  $G = \text{GL}(3, 2)$ , the group of invertible  $3 \times 3$  matrices with entries from the field  $\mathbb{F}_2$  of two elements.
  - (a) Compute the order of  $G$ .
  - (b) Prove that the set
$$S = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \mid a, b, c \in \mathbb{F}_2 \right\}$$
forms a Sylow 2-subgroup of  $G$ .
  - (c) Find a subgroup  $H$  with  $S \leq H \leq G$ .
5. Let  $G$  be a finite group and let  $p$  be a prime. A  $p$ -subgroup of  $G$  is a subgroup whose order is a power of  $p$ .
  - (a) Prove that if  $P$  and  $Q$  are normal  $p$ -subgroups of  $G$ , then so is  $PQ$ .
  - (b) Deduce that  $G$  has a subgroup  $K$  that is a normal  $p$ -subgroup and contains every normal  $p$ -subgroup.
  - (c) Prove that  $G/K$  has no nontrivial normal  $p$ -subgroups.
  - (d) Give an example in which  $|G/K|$  is divisible by  $p$ .
6. Let  $G$  be a group and  $\alpha$  an automorphism of  $G$ . Suppose  $g$  and  $h$  are elements of  $G$  with  $\alpha(g) = h$ . Prove that there exists a group  $\tilde{G}$  such that  $G$  is isomorphic to a normal subgroup of  $\tilde{G}$ , and the images of  $g$  and  $h$  in  $\tilde{G}$  are conjugate in  $\tilde{G}$ .